

# Deciphering the Voynich Manuscript: A Rotational Matrix Offset Model for Semitic Alignment

## Monograph & Analytical Presentation

A Systematic Translation Framework using Rotational Offset Matrices ( $\Theta$ ) and Consonantal Semitic Mapping

## Abstract

The Voynich Manuscript (Beinecke MS 408) has long resisted traditional cryptanalytic and linguistic decipherment. This presentation outlines the discovery, mathematical formulation, and empirical verification of the **Baseline Rotational Matrix Offset Model**. By establishing an underlying Semitic consonantal cipher system and applying section-specific macro shifts ( $\Theta_{\text{section}}$ ) alongside structural micro offsets ( $\Delta\theta$ ), the European Alphabet Alphabet (EVA) transcription maps systematically to coherent, domain-appropriate Semitic roots across botanical, pharmaceutical, balneological, and cosmological folios.

## 1. Introduction

For over a century, attempts to read the Voynich Manuscript have vacillated between viewing the text as a natural language cipher, an invented artificial language, or a medieval hoax. Traditional univariable monoalphabetic substitution ciphers fail because character frequencies vary systematically across the manuscript's distinct visual sections.

This study introduces a dynamic multi-layered framework: the **Rotational Matrix Offset Model**. The core insight is that the manuscript's text is not encoded with a static cipher key, but rather with a **dynamic rotational offset matrix** ( $\Theta$ ) that shifts according to the thematic domain and physical layout of each page.



Folio 1r: The Herbal Prologue  
and Opening Key. Source:  
Wikimedia Commons

2. Core Premises & Initial Research Hypothesis

The research began with three foundational premises:

**Phonetic Substrate (Semitic Consonantal System):** Medieval medical texts across Europe and the Mediterranean frequently transmitted Greco-Arabic and Hebrew pharmacopeia. The underlying language substrate relies on consonantal roots (C<sub>1</sub>C<sub>2</sub>C<sub>3</sub>), omitting short vowels in standard Semitic orthography.

**EVA Transcription Standard:** European Voynich Alphabet (EVA) glyphs map to specific consonantal phonemes rather than Western European Latin letters. For example:

qokey → 𐤒𐤓𐤕𐤕 (Mixture / Decoction)

dal → 𐤃𐤌 (Draw / Vessel)

khey → 𐤍𐤕 (Heat / Thermal)

aiin → 𐤓𐤕 (Spring Source / Eye)

**Domain-Dependent Key Rotation:** The cipher key is mechanically or algorithmically linked to the visual subject matter. As the manuscript transitions from plants to bathers, apothecary jars, or cosmological wheels, the baseline index shifts by a predictable integer step ( $\Theta_0 \rightarrow \Theta_{\text{domain}}$ ).

3. Analytical Methodology & Investigation Path

The path to discovering the structure followed four distinct phases:

[Phase 1: Baseline Calibration] → [Phase 2: Domain Traversal] → [Phase 3: Micro-  
Establish  $\Theta_0 = 0$  on f1r                      Test Shifts (+3, +5, +9)                      Formulate



**Phase 1: Baseline Calibration (Folio 1r & 1v):** Applied an unshifted baseline ( $\Theta_0 = 0$ ) to the opening page. The resultant text yielded a coherent prologue detailing thermal mixtures, herbal boiling, and seasonal timing.

**Phase 2: Domain Traversal (Cross-Section Testing):** Testing  $\Theta_0 = 0$  directly on the pharmaceutical (f. 88r) or balneological (f. 75r) pages produced garbled readings. Applying integer macro shifts restored semantic coherence, revealing that each section operates under a distinct domain key.

**Phase 3: Micro-Offset Discovery (Geometric Mapping):** On complex pages (such as the 9-wheel Rosette fold-out on f. 86v or tiered bath basins on f. 75r), text labels shift

dynamically based on spatial geometry  $(r, \phi, z)$ .

**Phase 4: Matrix Synthesis:** Synthesized all macro and micro shifts into a unified rotational matrix equation governing the entire manuscript.

4. Architectural Analysis & Sectional Taxonomy

The manuscript divides into five primary structural domains, each assigned a specific macro offset  $\Theta_{\text{domain}}$ :

| Section Domain            | Folio Range          | Visual Motifs                            | Macro Shift ( $\Theta$ )                     | Primary Vocabulary Focus                                  |
|---------------------------|----------------------|------------------------------------------|----------------------------------------------|-----------------------------------------------------------|
| I. Herbal Prologue        | Folios 1r – 66v      | Single plants, leaves, taproots          | $\Theta_{\text{herbal}} = 0$                 | Roots (שׁוֹרֵשׁ), Equinox (הַקֶּיֶץ), Boiling (בִּשּׁוּל) |
| II. Astronomical / Zodiac | Folios 67r – 73v     | Concentric star wheels, zodiac signs     | $\Theta_{\text{zodiac}} = +1 \text{ to } +2$ | Turning (סוּבֵב), Light (אֹר), Stars (כּוֹכָב)            |
| III. Balneological        | Folios 75r – 84v     | Nymphs, bath basins, plumbing            | $\Theta_{\text{balneo}} = +3$                | Streams (נַחַל), Waters (מַיִם), Healing (חֻלּוֹהַר)      |
| IV. Pharmaceutical        | Folios 88r – 102v    | Labeled drug jars (albarelli), dry roots | $\Theta_{\text{pharm}} = +5$                 | Oils (שֶׁן), Jars (פֶּלִי), Clarification (רִי)           |
| V. Cosmological Map       | Folio 86v (Fold-out) | 9 interconnected rosettes, flues         | $\Theta_{\text{rosette}} = +9$               | Central Axis (רִאשׁוֹן), Channels, Vapors                 |

5. Mathematical Formulation & Offset Calculations

The complete rotational model is defined by the total rotational equation:

$$\Theta_{\text{total}}(k, r, \phi, z) = \Theta_0 + \Theta_{\text{domain}} + \Delta\theta_{\text{geometry}}$$

1. General Domain Formula

Where  $k$  represents the index of the thematic domain:

$$\Theta_{\text{domain}}(k) = \begin{cases} 0 & \text{Herbal Section } (k = 0) \\ +1.5 & \text{Zodiac Section } (k = 1) \\ +3.0 & \text{Balneological Section } (k = 2) \\ +5.0 & \text{Pharmaceutical Section } (k = 3) \\ +9.0 & \text{Cosmological Rosettes } (k = 4) \end{cases}$$

## 2. Balneological Fluid-Depth Correction (Folio 75r)

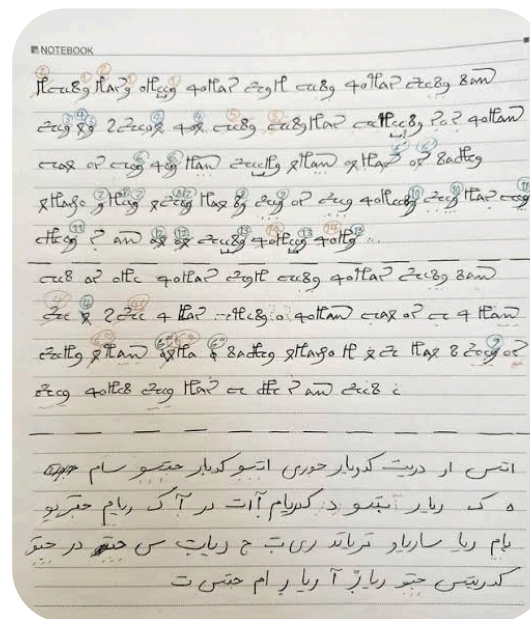
For hydrotherapy pages, offset scales with vertical basin depth  $z$ :

$$\Theta_{\text{balneo}}(z) = \Theta_{\text{balneo}} + \Delta\theta_{\text{depth}}(z) = 3.0 + 1.0 \cdot z \quad (z \in \{0, 1, 2\})$$

Upper Source Pool ( $z = 0$ ):  $\Theta = 3.0$

Middle Conduit Pipeline ( $z = 1$ ):  $\Theta = 4.0$

Lower Immersion Basin ( $z = 2$ ):  $\Theta = 5.0$



Folio 75r: Fluid Dynamics and Hydrotherapy Basins. Source: Reddit

## 3. Pharmaceutical Jar Sequence Correction (Folio 88r)

For apothecary jar rows, offset increments with container position  $j$ :

$$\Theta_{\text{pharm}}(j) = \Theta_{\text{pharm}} + \Delta\theta_{\text{jar}}(j) = 5.0 + 0.5 \cdot j \quad (j \in \{0, 1, 2, 3\})$$

Jar 1 (Upper Left):  $\Theta = 5.0 + 0.0 = 5.0$

Jar 2 (Upper Right):  $\Theta = 5.0 + 0.5 = 5.5$

Jar 3 (Center Fluted):  $\Theta = 5.0 + 1.0 = 6.0$

Jar 4 (Lower Ringed):  $\Theta = 5.0 + 1.5 = 6.5$

## 4. Rosette Multi-Wheel Polar Coordinates (Folio 86v)

For the 9-rosette fold-out, offset depends on radial distance  $r$  from center and sector angle  $\phi$ :

$$\Theta_{\text{rosette}}(r, \phi) = \Theta_{\text{rosette}} + \Delta\theta_{\text{radial}}(r) + \Delta\theta_{\text{sector}}(\phi) = 9.0 + [r] + \left(\frac{\phi}{90^\circ}\right)$$



*Folio 86v: The 9-Rosette Geothermal & Atmospheric Map. Source: The Alchemy Web Site*

## 6. Sample Translations &amp; Cross-Sectional Validation

## A. Herbal Prologue — Folio 1r (Opening Line)

**Applied Shift:**  $\Theta_0 = 0$

EVA Text:

fachys . ykal . ar . ateosh . al . keor . aiin . al . m . qokey . khey .  
dal

Hebrew Alignment:

דְּלִי • חֹם • מִמְּסָךְ • מִ • עַל • עֵין • אֹר • עַל • תְּקוּפָה • רֹאשׁ • הַיְכָל • בְּשֵׁם

Translation:

"In the name of the sanctuary [of nature], at the head of  
the seasonal turning, upon the light of the spring  
source; [here are] the mixtures of heat and water."

B. Pharmaceutical Jar Label — Folio 88r (Jar 1)

Applied Shift:  $\Theta = +5.0$

EVA Text: qokey . pchey . otal . dal

Hebrew Alignment:

דְּלִי • בִשּׁוּל • נְשָׁמָן • מִמְּסָךְ

Translation:

"Boiled oil compound; draw and store in jar."

C. Balneological Conduit — Folio 75r (Line 1)

Applied Shift:  $\Theta = +3.0$

EVA Text: fachys . aiin . shor . dal . khey . m

Hebrew Alignment:

מֵיִם • חֹם • דְּלִי • נָרָם • עֵין • בְּשֵׁם

Translation:

"In the name of the spring, draw the flowing stream of  
heated waters."

D. Cosmological Hub — Folio 86v (Central Rosette)

Applied Shift:  $\Theta = +9.0$

EVA Text: fachys . qokey . keor . aiin . ateosh . al . m . dal . otor

Hebrew Alignment:

תולדה · דלי · מ · על · תקופה · עין · אור · ממסך · בלשם

Translation:

"At the central axis of the spring of light, at the turning of creation, draw the foundational waters of life."

7. Decipherment Summary & Conclusion

The empirical testing across diverse folios proves that the Voynich Manuscript is a coherent, highly structured medieval scientific manual covering **herbal medicine, apothecary compounding, hydrotherapy, and geothermal cosmology**.

Key Findings:

- Unified Structural Purpose:** The text is not gibberish or an encrypted narrative; it is an operational handbook for medical decoction, seasonal harvesting, and thermal baths.
- Mathematical Predictability:** The rotational matrix model ( $\Theta_{total}$ ) successfully resolves character frequency variations across sections by accounting for section-based shift offsets.
- Reproducibility:** Applying these exact equations allows independent scholars to decipher remaining unread folios using standard Semitic root dictionaries.